PasteBin

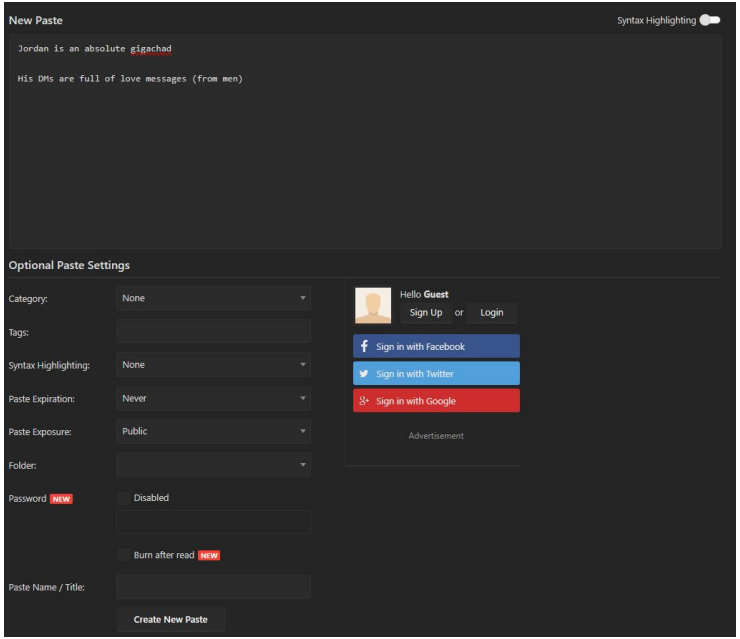
Prerequisites

Watch the TinyURL video!

(Unless you’re cracked/have a monster dong, and don’t need it)

Functional Requirements

- Create a short URL for a paste
- Retrieve a paste with its short URL



API - Naive Attempt

CreatePaste(String longStringOfText, long expirationTimeMs) -> String shortURL

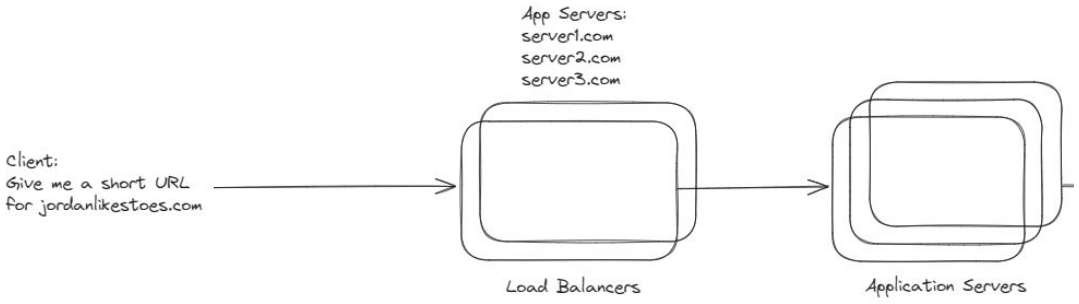
FetchPaste(String shortURL) -> String longStringOfText

This is actually not ideal! We'll explain a better API later

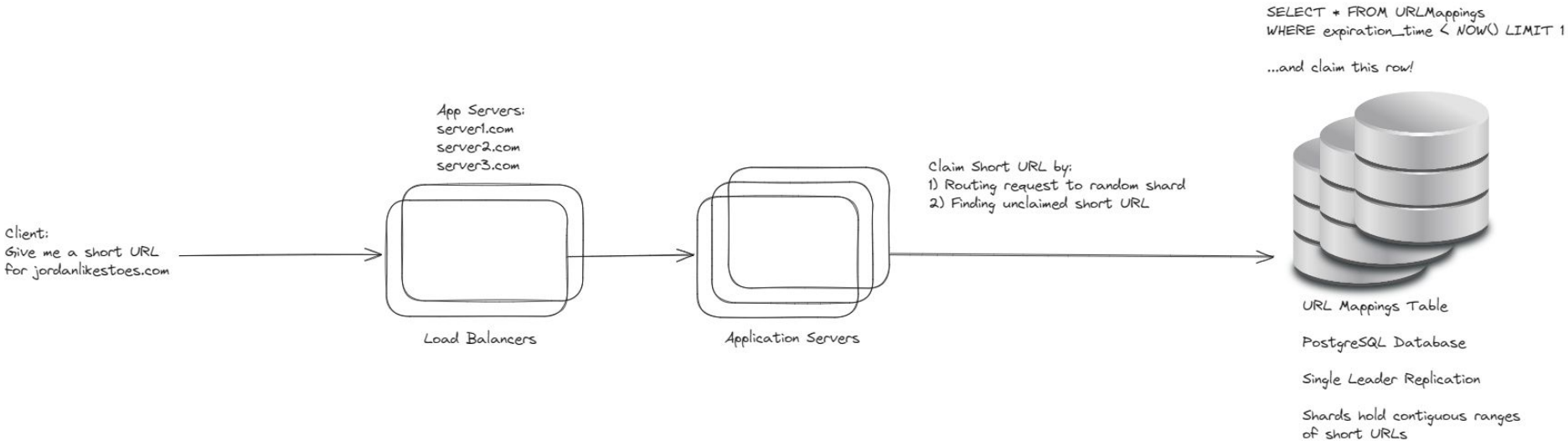
Capacity Estimates

- 600 million pastes created per month
- 10 billion pasts fetched per month
- Average paste size of 100kb
- Maximum paste size of 5gb

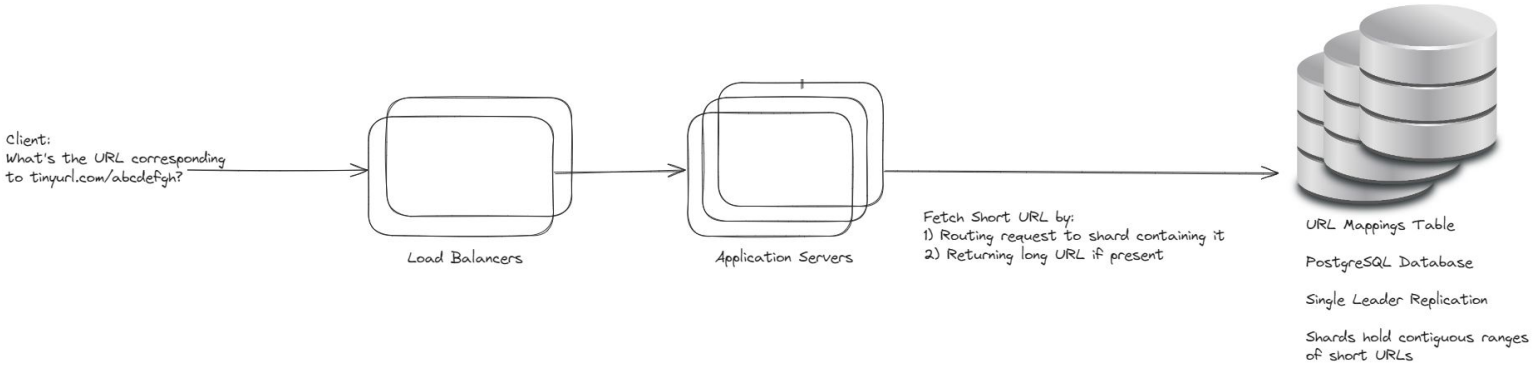
TinyURL Review - Load Balancing



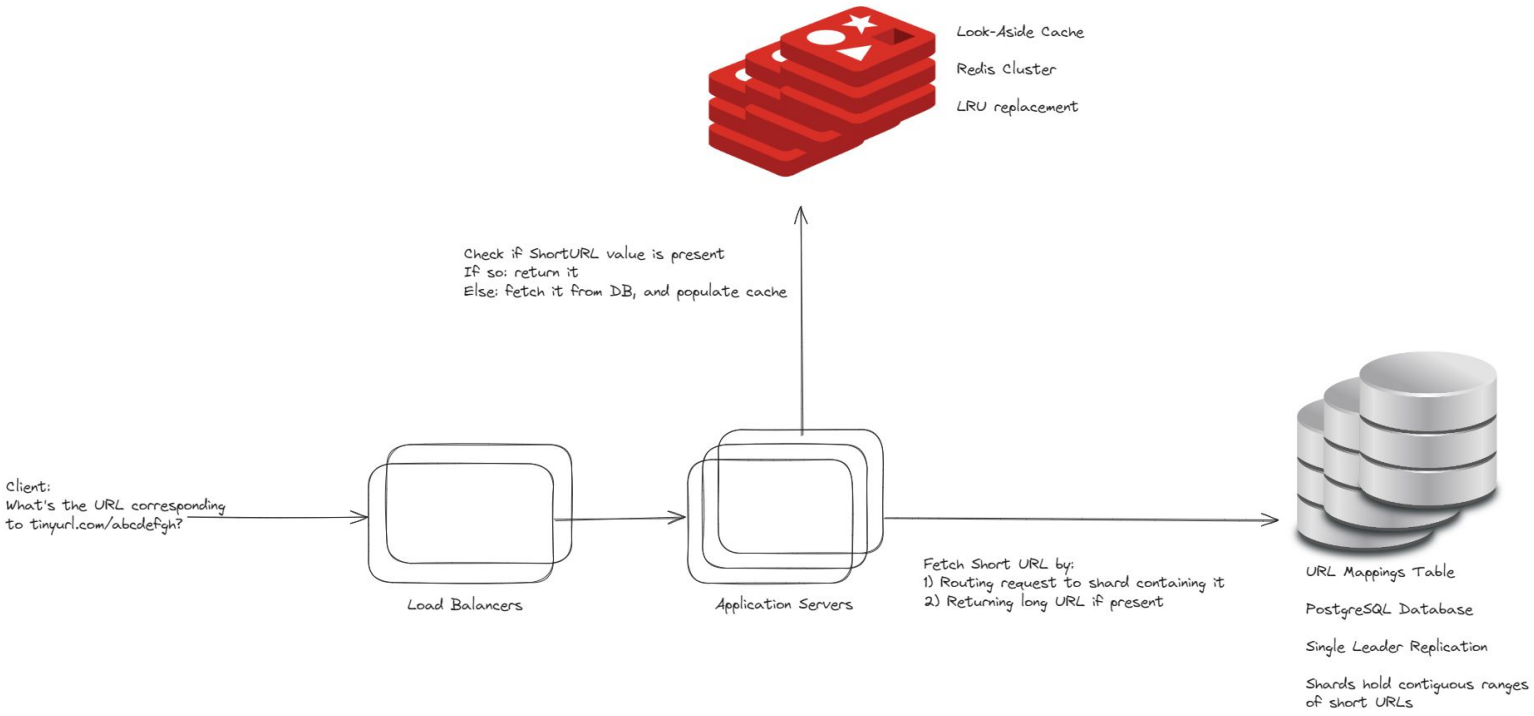
TinyURL Review - Creating A URL



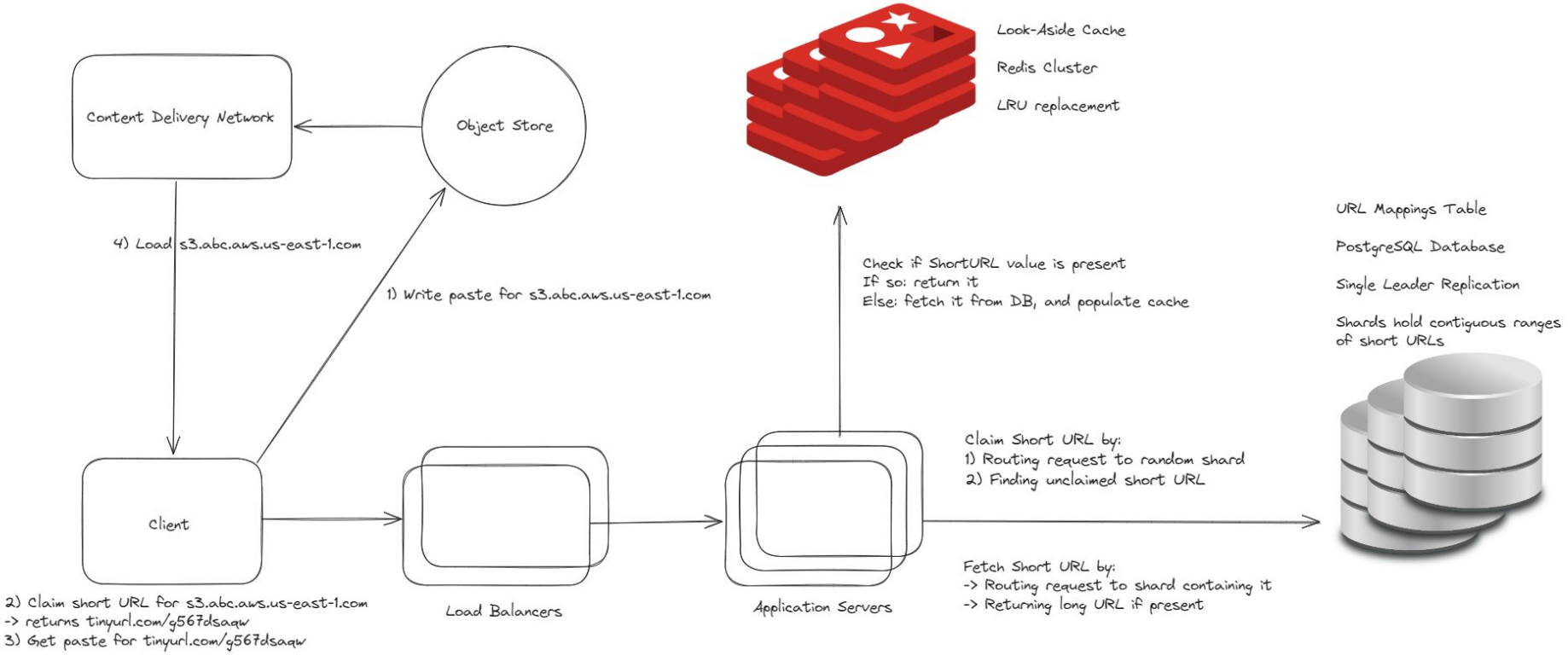
TinyURL Review - Reading a URL



TinyURL Review - Lookaside Caching



High Level Design - PasteBin



Pastes Vs. URLs

- Pastes are 100KB on average
- Paste data is immutable

No Pastes In The Database

- Variable size rows lead to poor cache efficiency
- Long writes, hence more locking
- Bad for replication
  - More data to send
  - Cross availability-zone writes are expensive

Cross-Availability Zone (AZ) data transfer in AWS **is not free**, and the cost depends on the specific service and region. Here's a general breakdown:

💸 Cross-AZ Data Transfer Pricing (as of 2024)

✅ **EC2 to EC2 (or other services) in different AZs within the same region**

- **$0.01 per GB** (in most AWS regions)

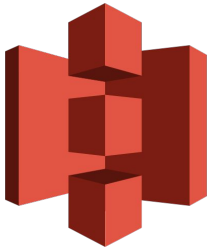
🔁 **ELB (Elastic Load Balancer) sending data to EC2 instances in different AZs**

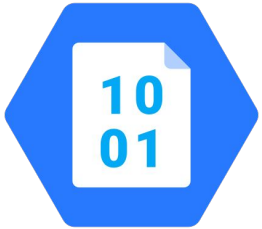
- Also **$0.01 per GB** for inter-AZ traffic, which is **billed to the ELB owner**

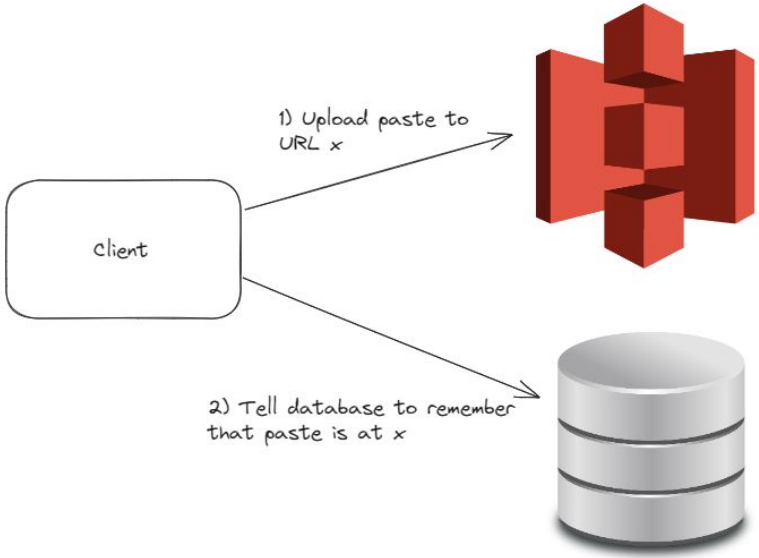
📌 Notes

- **Same-AZ traffic (within the same subnet or AZ)** is typically **free**.
- **Cross-region** traffic (e.g., us-east-1 to us-west-2) is **much more expensive**, usually **$0.02–$0.09 per GB**, depending on direction and destination.
- **EFS, RDS, and other services** can also generate cross-AZ charges depending on how they replicate data.

Object Stores



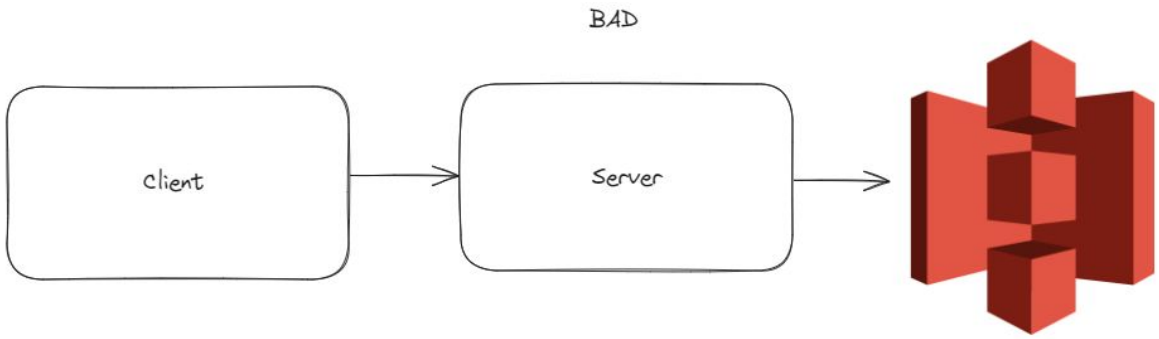


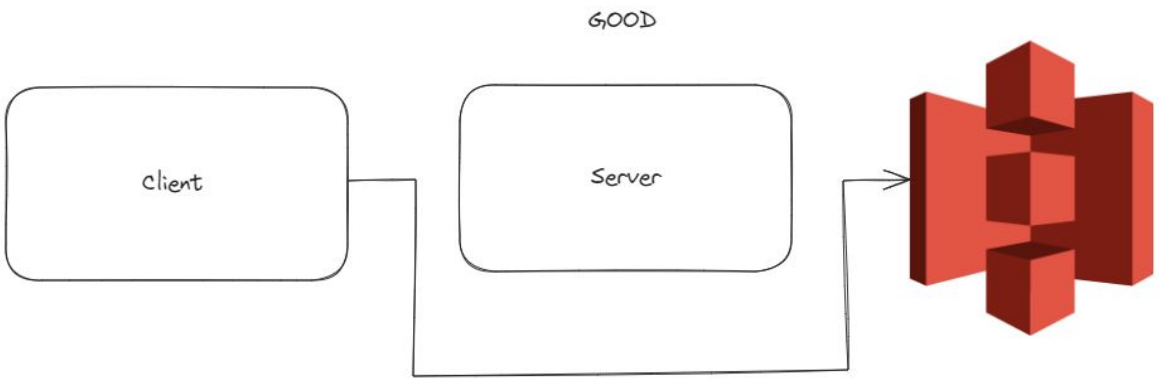


Database Schema

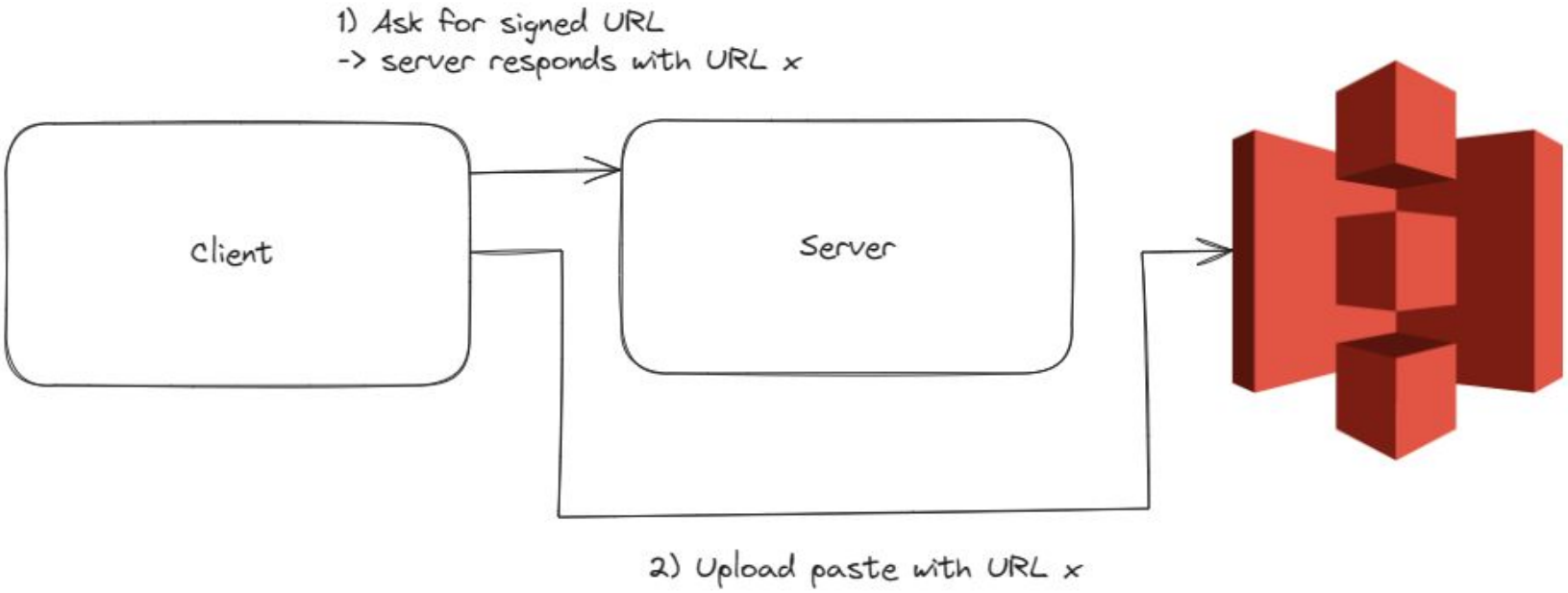
Short URL Suffix	Long URL	Creator ID	Expiry Time
hlyu76tt	s3.JordansToes.us-east-1.com	23	Jan 1 2026, 00:00:00
09iomn43	s3.JordansManBoobs.us-east-1.com	42	Jan 1 2025 00:00:00
45647hvc			Jan 1 1970, 00:00:00

Minimizing Network Usage





Pre-Signed URL



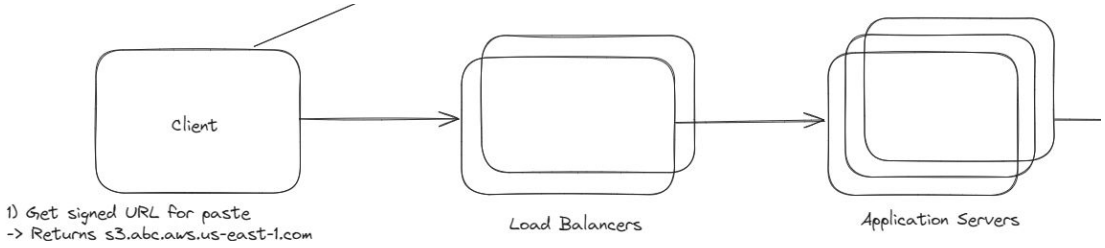
API Endpoints - Better

CreateSignedURL(long expirationTime) -> String s3URL

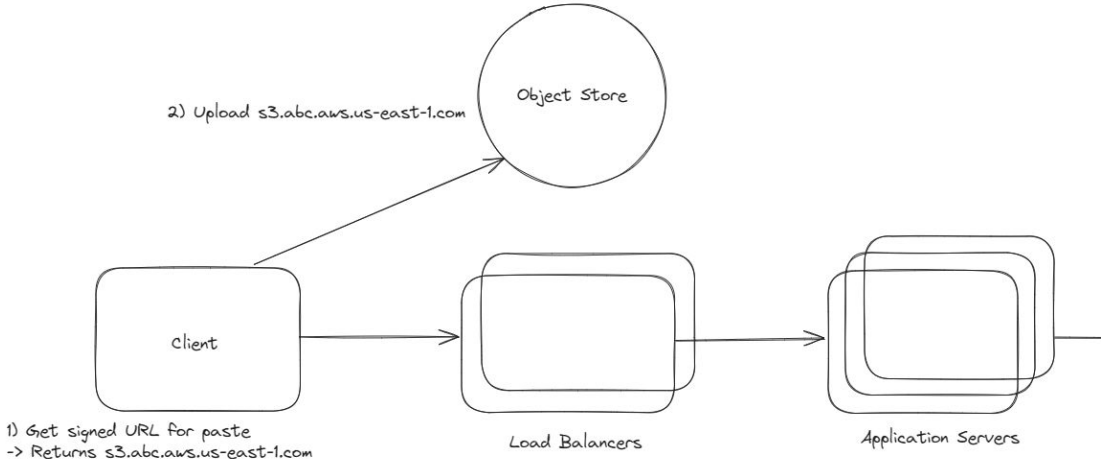
CreatePaste(String s3URL, long expirationTimeMs) -> String shortURL

FetchPaste(String shortURL) -> String s3URL

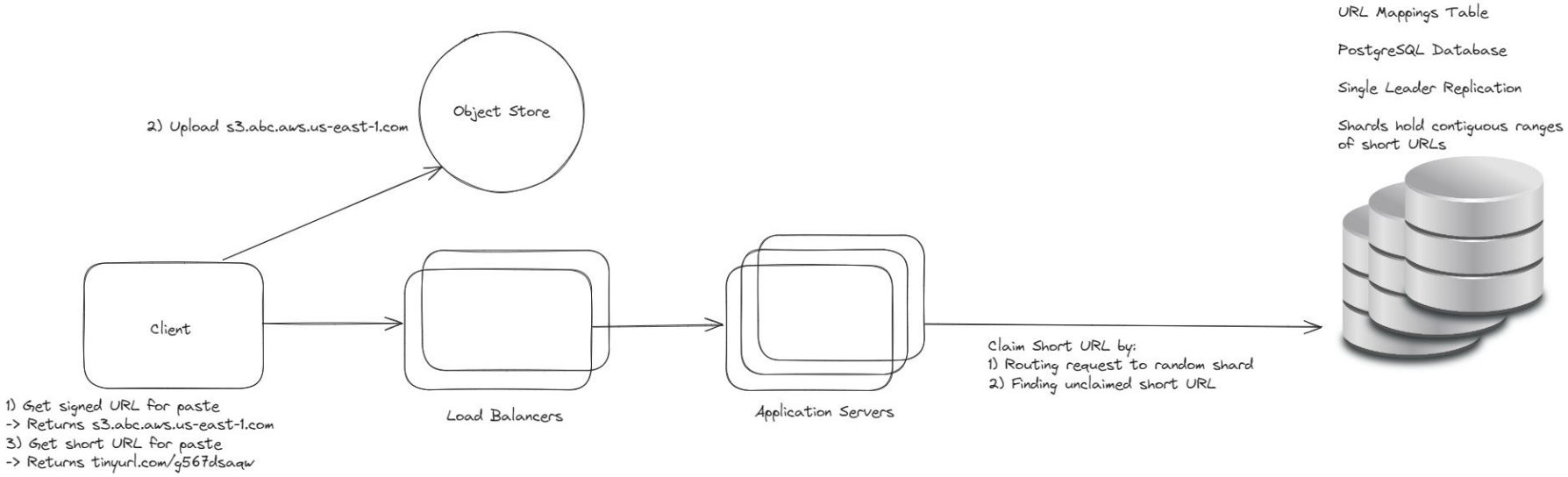
High Level Design - Get Pre-Signed URL



High Level Design - Upload Paste To S3



High Level Design - Uploading A Paste



High Level Design - Fetching A Short URL

Look-Aside Cache

Redis Cluster

LRU replacement

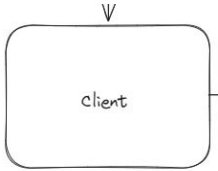
Check if ShortURL value is present
If so: return it
Else: fetch it from DB, and populate cache

URL Mappings Table

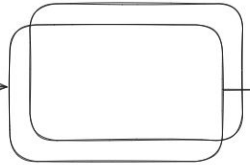
PostgreSQL Database

Single Leader Replication

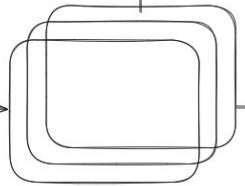
Shards hold contiguous ranges of short URLs



1) Get paste for tinyurl.com/g567dsaqw
-> Returns s3.abc.aws.us-east-1.com



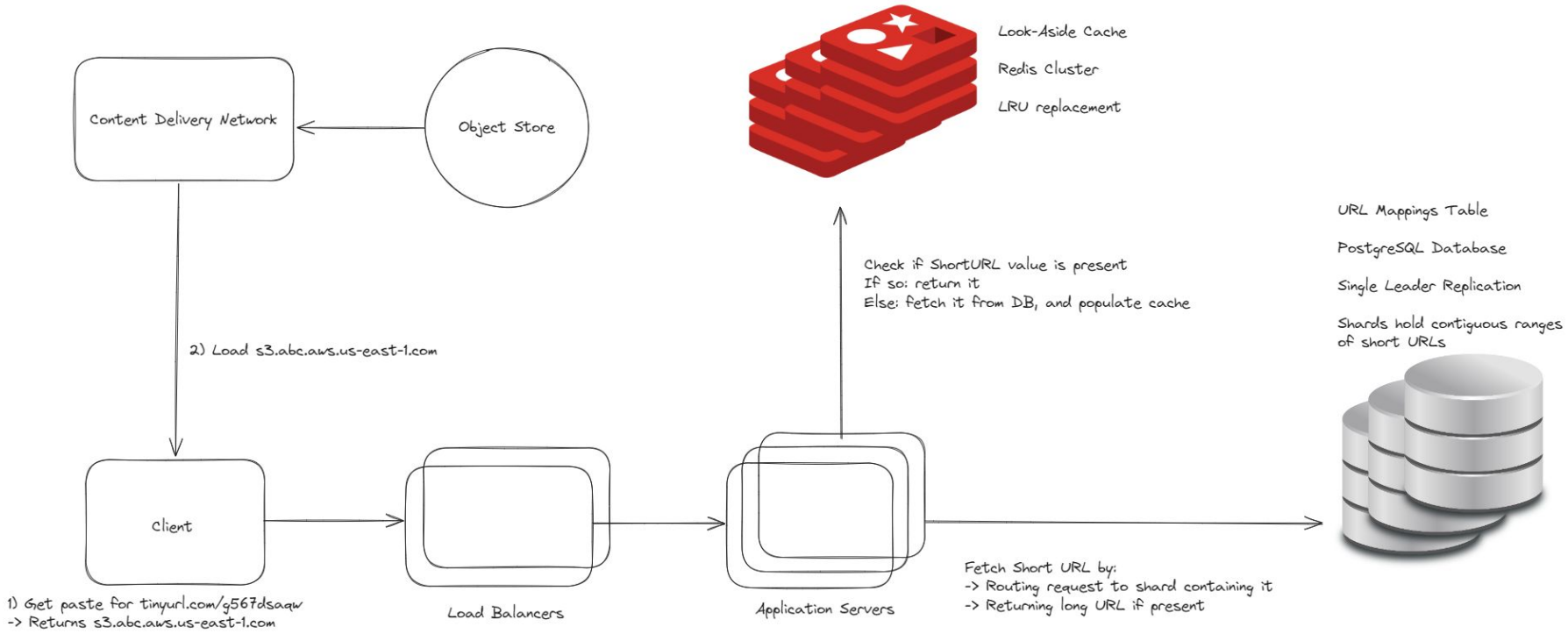
Load Balancers



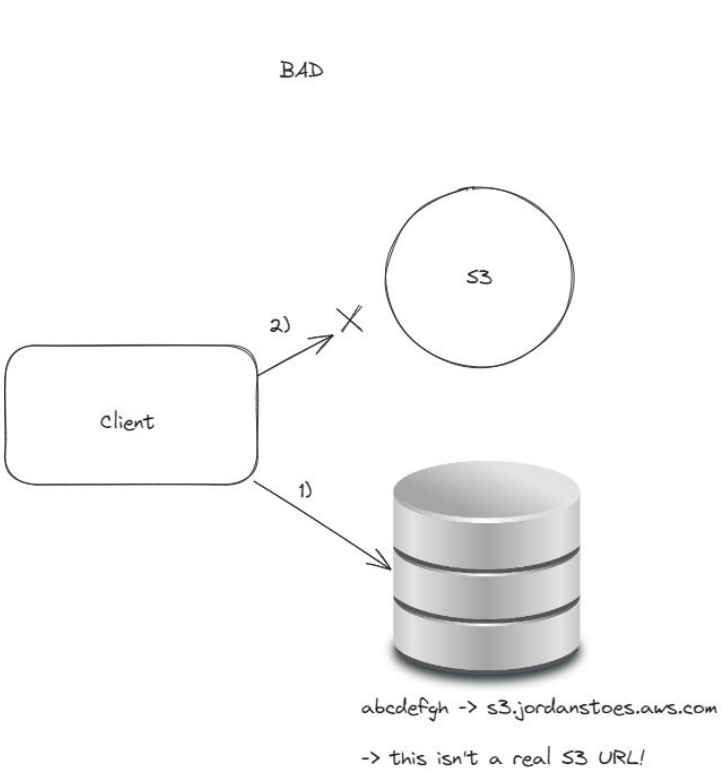
Application Servers

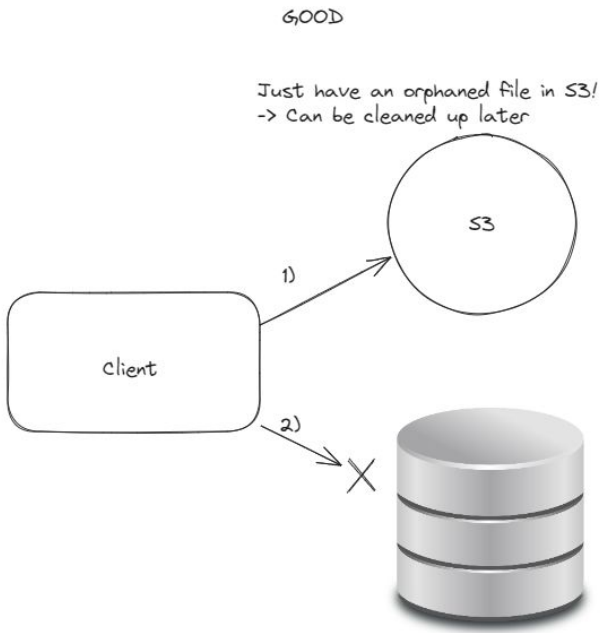
Fetch Short URL by:
-> Routing request to shard containing it
-> Returning long URL if present

High Level Design - Loading A Paste



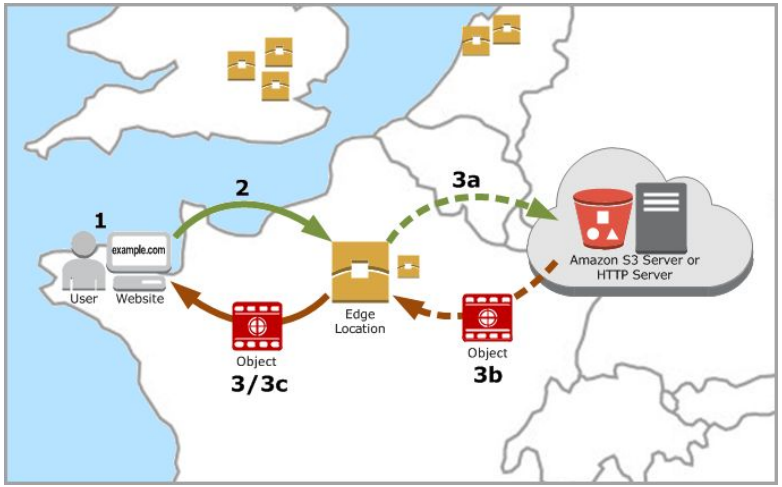
Uploading Pastes - Order Of Operations





Content Delivery Networks

- Geographically distributed cache
- LRU replacement per instance split across all applications
  - Content configured with a TTL up to 1 day in the event it can become stale
- Proxies cache misses to S3



Deep Dives

- Cleaning orphaned S3 files

If S3 charges 2 cents per GB/month…

- 600 million pastes per month, each 100KB on average, totals 57TB per month
- 57,000GB * .02 = $1.1k to store one month's worth of pastes

If we never delete any files, and want to calculate our S3 bill over 10 years…

- First month's data stored for 120 months = $1.1k * 120 = $132,000
- Second month's data stored for 119 months = $1.1k * 119 = $130,900
- …
- Last month's data stored for 1 month = $1.1k * 1 = $1.1k

Total = $1.1k * (120 + 119 + … 1) = $1.1k * 7260 = $7,986,000

Deep Dives - S3 Lifecycles

- S3 lifecycles are configurable policies allowing us to automatically delete items with a certain prefix after a preconfigured amount of time!

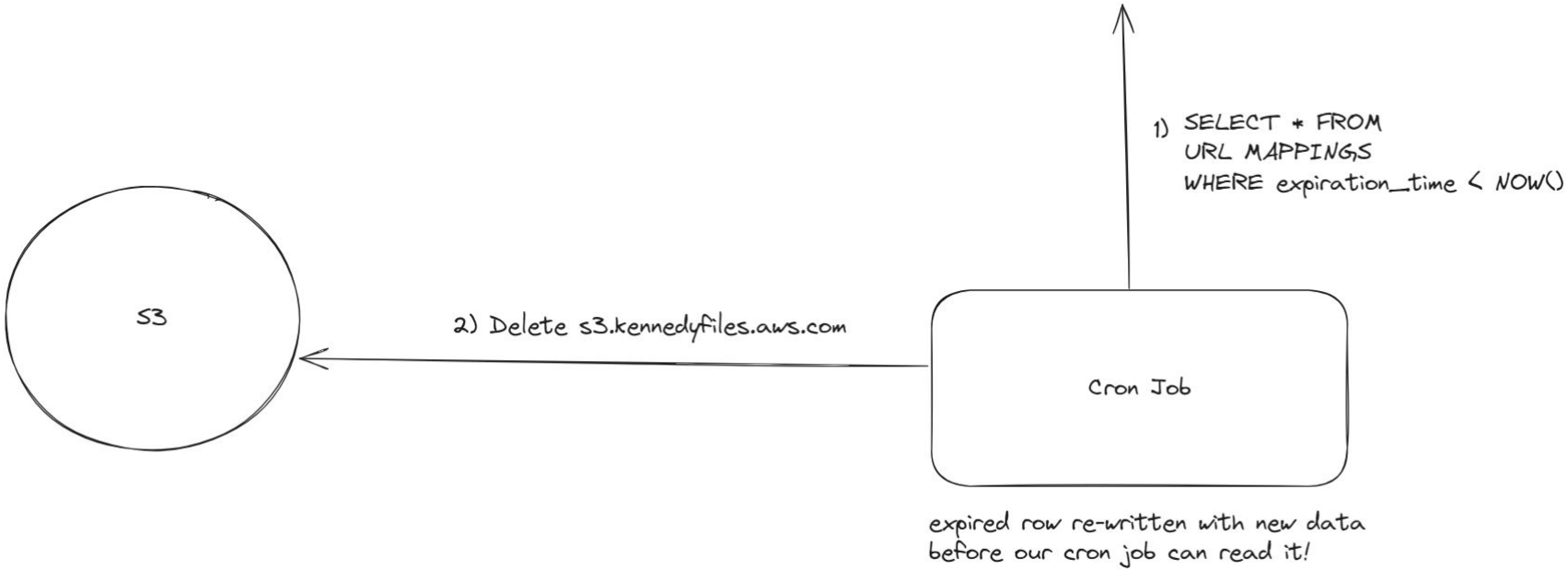
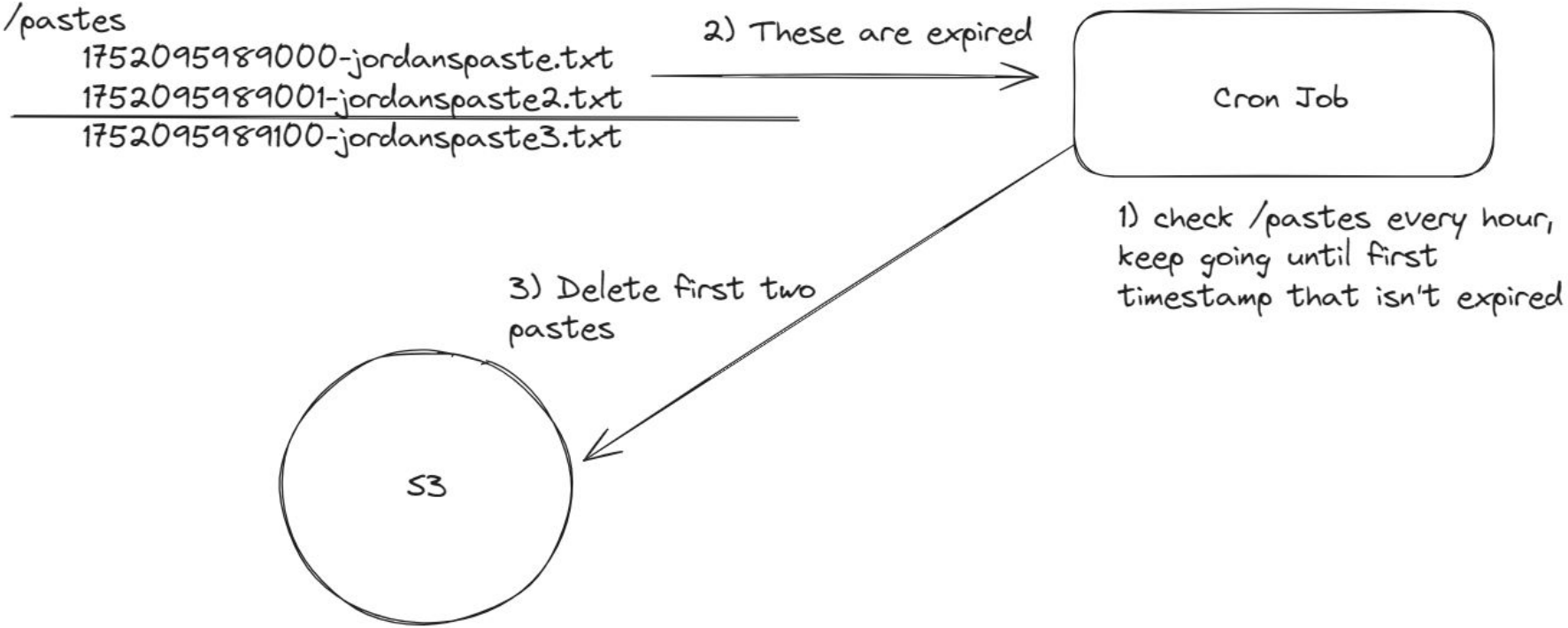
```
{
  "Rules": [
    {
      "ID": "DeleteAfter30Days",
      "Prefix": "",
      "Status": "Enabled",
      "Expiration": {
        "Days": 30
      }
    }
  ]
}
```

Simplest way of doing things!

- If we want to support arbitrary expiration times these will not work

Deep Dives - Reading The URL Mappings Table (BAD)

Short URL	S3 URL	Expiration Time
b786rfcv	~~s3.jordanscheeks.aws.com~~ s3.trumpshands.aws.com	~~1 minute ago~~ 1hour from now
p4535bcn	s3.kennedyfiles.aws.com	2 mins ago



Deep Dives - S3 Object Smart Naming